

Franziska Hoerbst

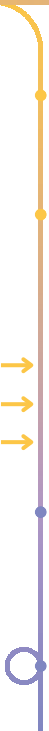
bouncing between research, art, activism & science communication

📍 On the move * 1998, Austria ✉️ franziska@hoerbst.net 🌐 franziskaaah.github.io

Research and education

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- 2024 – 2025**
Postdoc
John Innes Centre
Computational Systems Biology
Statistical pitfalls in the sciences and what we can do to resolve them
 - 2020 – 2024**
PhD
John Innes Centre & University of East Anglia
Computational Systems Biology
Doctoral Thesis supervised by Richard Morris
Bayesian Inference in the Analysis of RNA-Seq Data
 - 2017 – 2020**
BSc
University of Vienna
Molecular Biology
Thesis in Computational Biophysics with Bojan Zagrovic at Max Perutz Labs
Exploring the Physicochemical Foundation behind Travelling Autologous Protein-mRNA Pairs in the Phloem of Plants

Science communication, workshops & teaching

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- 2022 – 2024**
UEA Biodiversity and Climate Action Network
Building a movement; developing and running workshops and campaigns
 - 2020 – 2022**
🌐 more here
5MinutenClimateChance
Climate science podcast & radio broadcast
Co-founder, host, journalistic researcher & producer
 - Austrian SDG Award 2020
 - Austrian Environmental Journalism Prize 2020
 - Top 5 Salto Awards 2021
 - 2022**
Norwich,
University teaching
Mathematics for Computer Scientists
Teaching assistant
 - 2018 – 2020**
Vienna, AT
ScienceCenter Network Vienna
Hands-on science communication
in a job role called 'explainer', which is essentially a workshop host, who is developing hands-on and tinkering activities for audiences of all ages and backgrounds

Research publications

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|-------------------------|--|
| 2022 | • <u>Exact Bayesian inference for the detection of graft-mobile transcripts from sequencing data</u> |
| Royal Society Interface | F Hoerbst, M Tomkins, S Gupta, F Apelt, J Kehr, F Kragler, RJ Morris |
| 2023 | • <u>baymobil: A Python package for detection of graft-mobile mRNA using exact Bayesian inference on RNA-Seq data</u> |
| Preprint | F Hoerbst, RJ Morris, M Tomkins |
| 2024 | • <u>A Closed-Form Solution to the 2-Sample Problem for Quantifying Changes in Gene Expression using Bayes Factors</u> |
| Preprint | F Hoerbst, GS Sidhu, M Tomkins, RJ Morris |
| 2024 | • <u>Re-analysis of mobile mRNA datasets highlights challenges in the detection of mobile transcripts from short-read RNA-Seq data</u> |
| Nature | P Paajanen, M Tomkins, F Hoerbst, R Veevers, M Heeney, HR Thomas, F Apelt, E Saplaoura, S Gupta, M Frank, DJ Walther, C Faulkner, J Kehr, F Kragler, RJ Morris |
| 2024 | • <u>Bayesian Inference in the Analysis of RNA-Seq Data</u> |
| Thesis | F Hoerbst |
| 2025 | • <u>On the overinterpretation of mass screening data – the example of mobile mRNA</u> |
| Trends in Plant Science | F Hoerbst, M Tomkins, RJ Morris |
| 2024 | • <u>A Bayesian framework for ranking genes based on their statistical evidence for differential expression</u> |
| Under review | F Hoerbst, GS Sidhu, T Omori, M Tomkins, RJ Morris |
| 2024 | • <u>What is a differentially expressed gene?</u> |
| Under review | F Hoerbst, GS Sidhu, M Tomkins, RJ Morris |

Further publications & engagement

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|---------------|--|
| 2024 | • <u>Let's 'do different': Planetary Citizen Education Proposal</u> |
| Working paper | E Honeybun-Arnolda, P Stephanides, F Hoerbst, N Longhurst, D Maguire, H Hoechner |
| since 2020 | • <u>5MinutenClimateChance</u> |
| Podcast | 100+ episodes filled with snack-bites of climate research |
| 2021 | • <u>A day trip through phloem</u> |
| Blog | Easy read research project description |
| 2023 | • <u>Life out there? Scanning plant space for new communication networks</u> |
| Blog | Easy read paper teaser |
| 2023 | • <u>Activism and Advocacy as a Researcher</u> |
| Panel | Panel discussion at the Annual Assembly of the Tyndall Centre for Climate Change Research |

The miscellaneous

Most recent	→	John Innes Foundation Prize for Excellence in Scientific Reserach
2025	•	<u>The Unjournal</u> Evaluating research; to make <i>rigorous</i> research more <i>impactful</i> and <i>impactful</i> research more <i>rigorous</i>
	•	<u>Movement Reserach Unit</u> Volunteering to do the research that helps movements win
2024	→	Best Presentation Award at MaxEnt24 <i>Maximum Entropy (MaxEnt)</i> is a valued, historical conference series in the field of Bayesian statistics where researchers from quantum physics and philosophy to economy come together once a year since to learn from each other about probability theory
2023	→	JIC Director Award A yearly award given away by Director Graham Moore to individuals to acknowledge and honour outstanding scientific as well as cultural contributions to the John Innes Centre
since 2022	•	UEA Biodiversity and Climate Action Network Particularly involved in the Education Working group, with the vision of changing all curricula towards engaging all students in concepts surrounding global citizenship
🌐 more here		
since 2022	•	JIC Student Voice Runing campaigns to support the John Innes student community to thrive
2021	•	Finalist DigiEduHack 2021 Climate Change Digital Education Hackathon
from 2012	•	Radio Orange 94.0, Freies Radio Freistadt, & Radius 106.6 Speaker and producer
	→	Student award by the Austrian Federal Ministry of Education and OE1 for best broadcast 2016
2017 & 2018		Coordination and execution of the charity radio marathons <i>4 Tage für den Sozialmarkt, 4 Tage für die Lebenshilfe</i>
2016	→	Regional Movie Award @YOUKI Film Festival
2008 – 2017		Several accolades for music (<i>We can discuss the concept of music competitions, especially for kids and teenagers</i>)

finds balance in

- Music and silence
- Nature and the outdoors
- Arts and crafts
- Tech and low-tech
- Exploring movement and movements

speaks

- German, English and Spanish
- Python fluently

loves speaking

- at international scientific conferences
- on panels (like previously on Women in STEM, climate activism as researchers, ...)
- at public engagement events (like e.g. Pint of science)
- with kind humans around a dinner table

Finds a lot of enthusiasm for

Life
The post carbon world
Bayesian Statistics
Learning
Neuroplasticity
Philosophy of science
Equality & Justice
Colours & Complex Systems
Cooking meals for others
Spreading joy and seeing trees upside down
Slowing down & degrowth
Information Theory
The changing climate
Bicycles & Biodiversity
Movement & Movements
The importance of cultural practices
Staring out of the window on ridiculously long train journeys
Complex systems
New things every day

Contact for references

Professor Richard J. Morris

Group leader at John Innes Centre

Primary PhD supervisor

Richard.Morris@jic.ac.uk

Dr Noel Longhurst

Lecturer in Energy & Climate Change, School of Environmental Sciences, UEA and founding member of UEA BCAN

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